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Mini Project Report on

“Smart Waste Segregation System”

Submitted in partial fulfillment of the requirements for the
First Semester of the Bachelor of Engineering Degree, towards the completion of the **Mini Project** under the **Innovation & Design Thinking Laboratory**,
Department of Basic Sciences.

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CERTIFICATE

This is to certify that the File Structures mini project entitled “Smart Waste Segregation System ” has been successfully carried out by Ankit Kumar Goswami (1CR25IS025), Chinmayi G.J (1CR25AI024), and Anjali S Gudami (1CR25IS024), Chetan Reddy(1CR25AI023) bonafide students of **CMR Institute of Technology**.

The project is submitted in partial fulfillment of the requirements for the First Semester of the Bachelor of Engineering Degree, towards the completion of the Mini Project under the **Innovation & Design Thinking Laboratory, Department of Basic Sciences**.

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. This File Structures mini project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

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ABSTRACT

The Smart Waste Segregation System is an automated solution designed to efficiently separate different types of waste such as wet, dry, metal, and plastic waste using sensors in intelligent technologies. The system helps reduce manual effort, improves recycling efficiency, and supports proper waste management practices. It uses components like moisture sensors, metal detectors, microcontrollers, and servo motors to identify and sort waste into separate bins automatically. Advanced versions may also use IoT and machine learning for monitoring and accurate classification. This project promotes environmental sustainability, reduces pollution, and contributes to cleaner and smarter cities by improving the waste disposal process. Smart Waste Segregation System : IoT Automation, Sensors Recycling, Microcontroller Arduino, Embedded System Smart City, Machine Learning, Metal Detection, Moisture Sensor.

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Chapter 1

INTRODUCTION

Rapid urbanization and increasing waste generation have made conventional waste management systems inefficient, leading to poor segregation, environmental pollution, pest infestation, and improper disposal of cigarette butts, which contain toxic chemicals and pose fire and health hazards. To address these challenges, the Smart Waste Segregation System with automation, and control mechanisms to automatically segregate waste, prevent pest breeding through automated pest control, and safely This system reduces manual intervention, improves hygiene, promotes recycling, and ensures environmentally responsible waste handling, making it suitable for public places, institutions, and smart city applications while contributing to sustainable and cleaner surroundings.

1.1 Brief history of waste segregation

Waste management practices have evolved from simple open dumping and manual disposal methods in early civilizations to organized collection systems during industrialization, driven by increasing urban populations and public health concerns. The introduction of waste segregation and recycling gained importance in the late 20th century with growing environmental awareness, while technological advancements led to the use of sensors, automation, and smart bins in modern waste management. Recent developments focus on integrated systems that not only segregate waste but also address associated issues such as reflecting a shift toward intelligent, sustainable, and smart city-oriented waste management solutions.

1.2 Modern waste segregation

Modern waste segregation utilizes smart technologies such as sensors, microcontrollers, IoT, and automated sorting mechanisms to efficiently separate waste at the source into biodegradable, recyclable, hazardous, and special categories .These systems reduce human intervention, improve recycling efficiency, and minimize environmental pollution by ensuring proper handling of waste. By integrating real-time monitoring, data analytics, and additional features and safety mechanisms, modern waste segregation plays a crucial role in maintaining hygiene, supporting sustainability, and enabling smart city waste management solutions.

Chapter 2

PROBLEM STATEMENT

2.1 Description

The absence of an efficient and automated waste management system leads to improper waste segregation, increased environmental pollution, in public and institutional areas. Existing manual systems are unhygienic, labor-intensive, and ineffective in managing mixed waste, resulting in poor recycling efficiency and health hazards. Hence, there is a need for a smart waste segregation system with ensure hygienic, sustainable, and environmentally safe waste disposal.

2.2 Challenge Statement

The key challenge is to design and implement a cost-effective, reliable, and automated system that can accurately segregate different types of waste. The system must operate with minimal human intervention, maintain hygiene, withstand varying environmental conditions, and be suitable for deployment in public and institutional spaces while ensuring sustainability and ease of maintenance.

Chapter 3

DESIGN

3.1 Design Thinking Process

3.1.1.Empathize:

Understand the problems faced by people due to mixed waste.

3.1.2.Define:

Define the problem as the need for an automated system to segregate waste, control pests, and safely collect cigarette butts to improve hygiene.

3.1.3.Ideate:

Generate ideas to use sensors, automation, mechanisms to create a smart waste management solution.

3.1.4.Prototype:

Develop a working model using sensors, microcontroller, motors, and compartments to demonstrate automatic segregation

3.1.5.Test:

Test the system in real conditions to check accuracy, reliability, and effectiveness in waste segregation.

3.2 Methodology

3.2.1Working Principle (Overview)

The system is designed to:

Automatically segregate waste (wet, dry, metal)

Detect pests (rats/insects) and activate control mechanisms

Collect cigarette butts separately to avoid fire risk and pollution

It uses sensors, microcontroller, motors, and pest control modules to perform all actions without human contact.

3.2.2.Components Used (Typical)

- 1.IR Sensor / Ultrasonic Sensor – Object detection
- 2.Moisture Sensor – Wet waste detection
- 3.Metal Sensor – Metal waste detection
- 4.Servo Motors – For opening flaps
- 5.Buzzer / Spray Module – Pest control
- 6.Microcontroller (Arduino / ESP32)
- 7.Separate compartments (Wet, Dry, Metal)

3.3.Working Procedure (Step-by-Step)

Step 1: Waste Detection

When waste is dropped into the inlet, an IR/Ultrasonic sensor detects the presence of waste and signals the microcontroller.

Step 2: Waste Type Identification

The system checks:

Moisture Sensor → detects wet waste

Metal Sensor → detects metal objects

Gas / Vision Sensor → identifies cigarette butts

If none → considered dry waste

Step 3: Segregation Mechanism

Based on sensor data, the microcontroller:

Rotates the servo motor

Opens the corresponding compartment flap

Drops the waste into the correct bin:

*Wet bin

*Dry bin

*Metal bin

Step 4 : System Reset

After waste disposal and pest control:

System resets

Ready for next waste input

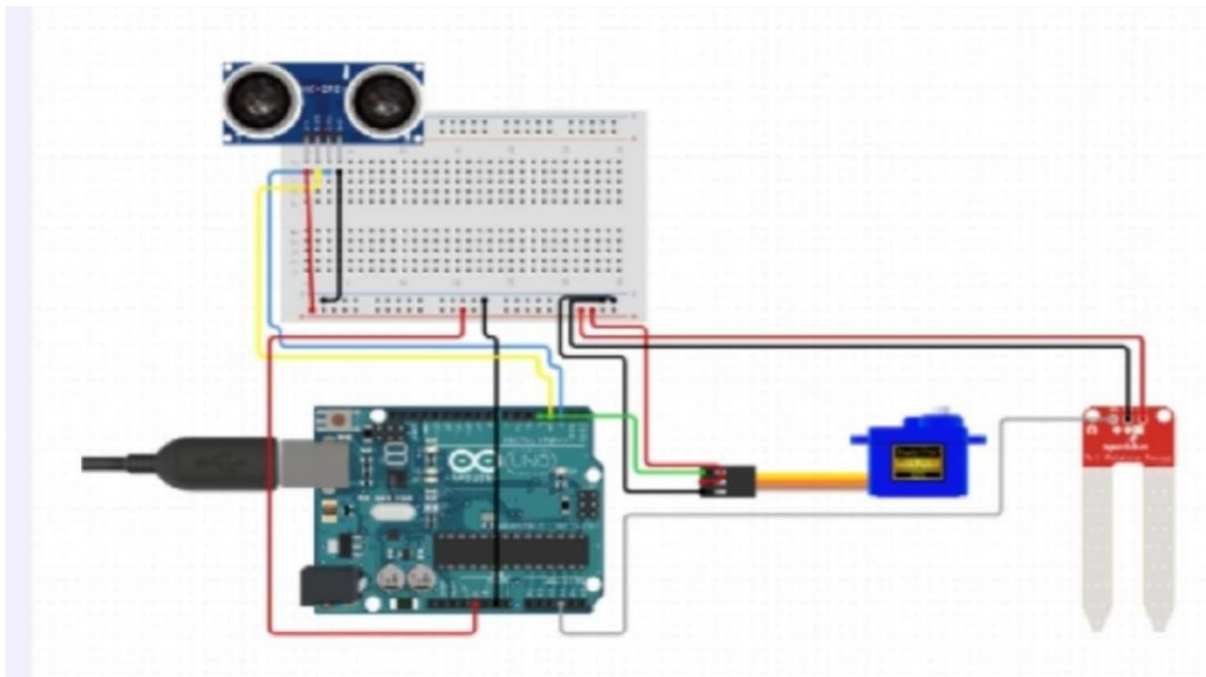
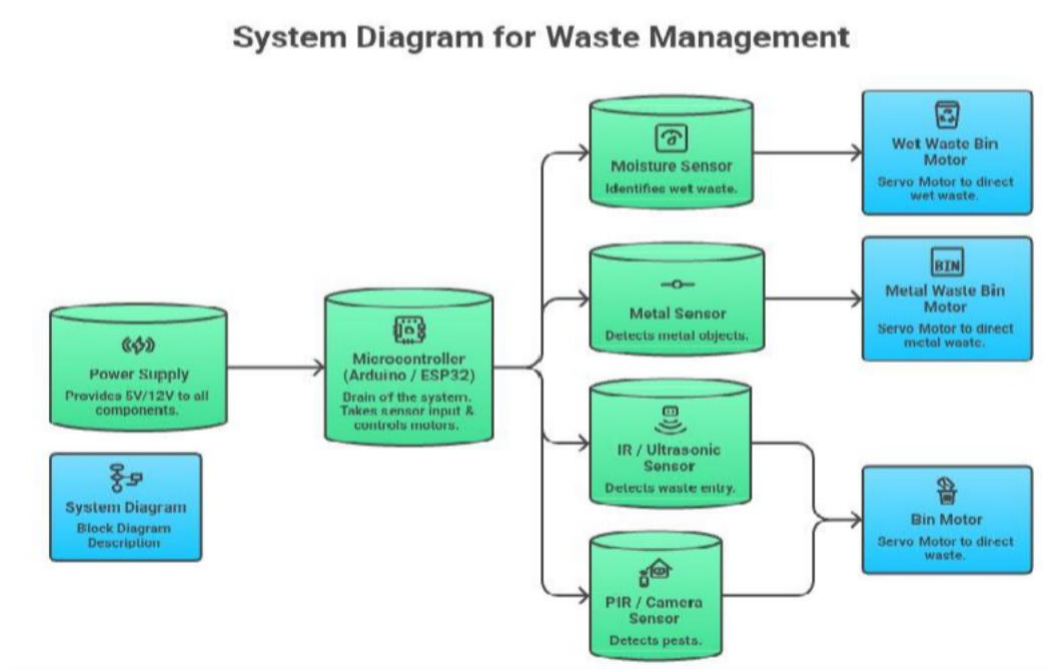


Fig 3.3: Circuit diagram of the smart waste segregation system

3.3.1 System Diagram



3.3.2. Block Diagram Description

- *Power Supply: Provides 5V/12V to all components.
- *Microcontroller: Brain of the system. Takes sensor input & controls motors.
- * IR/Ultrasonic Sensor: Detects waste entry.
- * Moisture Sensor: Identifies wet waste.
- * Metal Sensor: Detects metal objects.
- * Servo Motors: Direct waste into correct compartment.
- * PIR / Camera Sensor: Detects pests.

3.3.3. Circuit Connection Diagram (Text Explanation)

- * Sensor Connections
- * Component are:
 1. Arduino Pin IR Sensor :D2
 2. Moisture Sensor :A0
 3. Metal Sensor :D3
 4. PIR Sensor :D5
 5. Servo Motor 1 (Wet) :D6
 6. Servo Motor 2 (Dry) :D7
 7. Servo Motor 3 (Metal) :D8
 8. Servo Motor 4 (Cigarette) :D9

Chapter 4

IMPLEMENTATION

4.1.Program code

```
#include <Arduino.h>
#include <NewPing.h>
#include <Servo.h>
// ----- PIN DEFINITIONS -----
#define TRIG_PIN 3
#define ECHO_PIN 2
#define SERVO_PIN 4
#define MOISTURE_PIN A3

// ----- OBJECTS -----
NewPing sonar(TRIG_PIN, ECHO_PIN, 30); // Max distance 30 cm
Servo wasteServo;

// ----- SERVO POSITIONS -----
const int DRY_POSITION = 20; // Dry waste side
const int WET_POSITION = 150; // Wet waste side

// ----- MOISTURE THRESHOLD -----
int moistureThreshold = 500; // Adjust after testing

void setup() {
  Serial.begin(9600);

  wasteServo.attach(SERVO_PIN);
  wasteServo.write(DRY_POSITION); // Default dry
  delay(500);
}

void loop() {
```

```
int distance = sonar.ping_cm();

if (distance > 0 && distance < 10) { // Waste detected
  Serial.println("Waste detected");

  int moistureValue = analogRead(MOISTURE_PIN);
  Serial.print("Moisture Value: ");
  Serial.println(moistureValue);

  if (moistureValue < moistureThreshold) {
    // ----- WET WASTE -----
    Serial.println("Wet Waste");
    wasteServo.write(WET_POSITION);
  }
  else {
    // ----- DRY WASTE -----
    Serial.println("Dry Waste");
    wasteServo.write(DRY_POSITION);
  }

  delay(1500); // Time to drop waste
  wasteServo.write(DRY_POSITION); // Reset position
  delay(2000);
}
}
```

Chapter 5

RESULT AND ANALYSIS

5.1 User Testing & Feedback

Sample:

Participants: 12 students and 2 faculty members.

Quantitative Results:

- - Scan time reduced to 1.4 seconds.
- - Queue time reduced from 10 minutes to 1 minute.
- - Accuracy: 100% during testing.

Qualitative Feedback:

- - Students: “Much faster than existing device.”
- - Faculty: “Live dashboard is extremely helpful.”
- - Admin: “No data mismatches during testing.”



Figure 5.1.1: Smart waste Segregation Prototype model

The smart waste segregation system uses sensors and a microcontroller to automatically detect and separate wet and dry waste through a motorized mechanism. Preventing fire hazards and environmental pollution. An automated pest control diffuser operates at preset intervals or based on conditions to repel mosquitoes and flies around the bin. The system delivers accurate waste segregation with minimal human involvement, reduces waste mixing, improves hygiene, and lowers pest-related health risks. Overall, it enhances cleanliness, promotes responsible disposal practices, and supports efficient and sustainable waste management in public and residential spaces.

Chapter 6

CONCLUSION & FUTURE WORK

The Smart Biometric Attendance System effectively addresses inefficiencies in traditional attendance methods. It enhances accuracy, reduces time wastage, and improves administrative transparency.

Future Work:

- AI-based facial recognition.
- Predictive absentee analytics.
- Automated parent notifications.
- Full ERP/LMS integration.

The Smart Waste Segregation System developed in this project provides an effective solution for automatic waste segregation and hygienic waste handling. However, the system can be further enhanced by integrating advanced technologies and additional smart features to improve efficiency, accuracy, and large-scale usability.

In the future, Artificial Intelligence (AI) and Machine Learning (ML) techniques can be incorporated to improve waste identification and classification. Instead of relying only on moisture and metal sensors, image processing with cameras and AI models can be used to recognize different categories of waste such as plastic, paper, glass, food waste, and hazardous materials with higher accuracy. This would make the system more intelligent and adaptable to real-world waste management requirements.

The system can also be upgraded with Internet of Things (IoT) technology for real-time monitoring and data collection. By connecting the smart bin to cloud platforms or mobile applications, users and municipal authorities can monitor the fill level of bins, track waste collection schedules, and receive alerts when bins are full. This will help in optimizing waste collection routes, reducing operational costs, and improving city cleanliness.

Another future improvement is the addition of renewable energy support such as solar panels to power the system. Solar-powered smart bins can reduce electricity consumption and make the system suitable for outdoor public environments, especially in smart city projects. Energy-efficient components and low-power microcontrollers can also be implemented to increase

sustainability.

The pest control mechanism can be enhanced further by integrating automatic disinfectant spraying and odor control systems. This will improve hygiene and reduce the spread of harmful bacteria and insects around waste disposal areas. Smart air-quality sensors can also be added to detect harmful gases released from waste and provide safety alerts.

Future versions of the system may include advanced recycling features such as compacting plastic waste, crushing cans, or separating biodegradable waste for compost generation. This would increase recycling efficiency and reduce landfill waste. Additionally, the cigarette butt collection mechanism can be improved with fire-detection and fire-extinguishing features to prevent accidental fires in public spaces.

The system can also be made more user-friendly by integrating voice guidance, display screens, or multilingual instructions for public use. Mobile app integration can allow users to access waste disposal information, recycling tips, and system status updates. Such features would encourage public participation in proper waste management practices.

Finally, the proposed system can be scaled and implemented in schools, hospitals, railway stations, shopping malls, and smart cities for efficient waste management. With further research and development, the Smart Waste Segregation System has the potential to become a fully automated, eco-friendly, and intelligent solution for sustainable waste management in the future.

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Annexures

Annexure A – User Feedback Forms

Annexure B – Iteration Notes

Annexure C – Team Roles